EB-JEPA for Clinical EEG

Self-Supervised Abnormality Detection — Honest Results

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Hackathon — June 2026

Why Self-Supervised EEG?

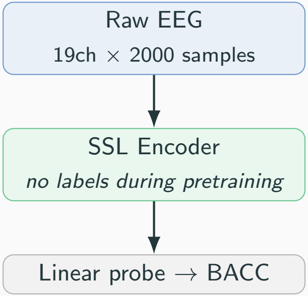
Clinical reality:

- Every EEG must be reviewed by a licensed neurologist
- Labels are scarce and expensive
- Unlabelled EEG data is abundant

Goal: learn a *transferable* EEG representation **without labels**

Benchmark — TUAB v3.0.1:

- Binary: normal / abnormal clinical EEG
- 19 channels, 200 Hz, 10 s windows
- 2 717 train / 276 eval (patient-disjoint)

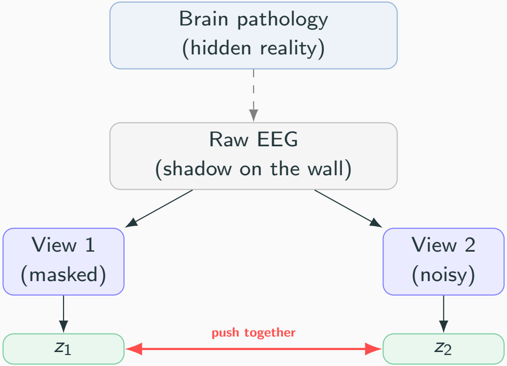


Plato's Cave — What SSL Actually Does

Prisoners in Plato's cave see only **shadows on the wall** — not the real objects casting them. They learn to *predict and recognise* patterns in the shadows without ever seeing the cause.

In our setting:

- The cave wall = raw EEG voltages
- The hidden reality = underlying brain pathology
- The SSL encoder = a prisoner learning that *two corrupted views of the same shadow are the same object*
- The linear probe = asking the prisoner to name what they see



The encoder never sees a neurologist's label. It learns structure from the signal alone.

EB-JEPA — Two-View Invariance + Corruption

Two-view SSL pipeline:

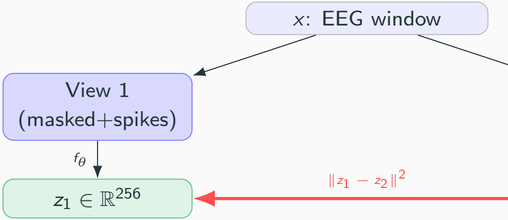
1. Take one EEG window; create two augmented views v_1, v_2
2. Encode both: $z_i = f_\theta(v_i) \in \mathbb{R}^{256}$
3. Minimise $\|z_1 - z_2\|^2$ + anti-collapse regulariser

Augmentations (standard): Gaussian noise, amplitude jitter, channel drop

Our design — Corruption augmentation:

- 20 % time steps **masked** to zero
- 20 % time steps injected with $\pm 6\sigma$ **spikes**
- Mimics real EEG artefacts; forces robust representations

Two objectives tested: VICReg ($V{+}C$ regulariser) and SIGReg (Epps-Pulley Gaussianity test)



Finding 1 — Corruption Augmentation: the Dominant Lever

Variant	$\mathrm{BACC}_{\mathrm{REC}}$	Δ
Base VICReg (no corruption)	0.796	—
+ Corruption (mask + spikes)	**0.819 ± .004**	+0.023
+ Spectral loss	0.795–0.813	≈0
Scaled encoder	0.805	−0.014
4× more data	0.812	−0.007

3-seed average (robust):
0.819 ± 0.004 — not a lucky seed

Why corruption helps:

- Standard augmentation (noise, jitter) creates views that are "too easy" to match
- Masking + spike injection forces the encoder to learn *content*, not artefact patterns
- Mirrors the real EEG noise distribution

What does NOT help:

- More encoder capacity → same plateau
- 4× pretraining data → no gain

⇒ The bottleneck is the *architecture*,

Finding 2 (Novel) — EEG Prefers Weak VICReg Invariance

VICReg loss:

$$\mathcal{L} = \lambda_{\text{inv}} \underbrace{\|z_1 - z_2\|^2}_{\text{invariance}} + 25\, V + 1\, C$$

Paper-correct: $\lambda_{\text{inv}} = 25$. Framework default: $\lambda_{\text{inv}} = 1$.

λ_{inv}	BACC_{REC}	Seeds
$\lambda = 1$ (default)	**0.819 ± .004**	3
$\lambda = 25$ (paper)	0.789 ± .005	3

Gap: 3 pp, $\approx 6\sigma$ — not noise

Why?

- With λ=1, anti-collapse ($V+C$) **dominates** invariance
- Encoder focuses on *spreading* representations across dimensions — more discriminative
- With λ=25, invariance dominates: encoder collapses toward view-similarity at the cost of feature diversity

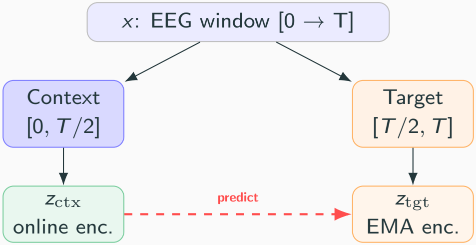
Ongoing ablation: $\lambda \in \{1, 5, 10, 25\}$ (capDinv5 / capDinv10 running now)

This is, to our knowledge, the first systematic evidence that EEG SSL requires weaker view-invariance than

NaiveJEPA — Temporal-Split Prediction, No Anti-Collapse Needed

Idea: split one window in time instead of augmenting.

- Context ($0 \to T/2$): online encoder $\to$ mean-pool
- Target ($T/2 \to T$): EMA encoder $\to$ mean-pool
- Loss: $\mathrm{MSE}(\hat{z}_{\mathrm{tgt}},\ z_{\mathrm{tgt}})$
- **No VICReg, no SIGReg** needed: context and target see *disjoint time spans* — collapse to a constant is architecturally impossible



Only 0.04 below VICReg + 3 loss coefficients, with 1 hyperparameter.

Method	$\mathrm{BACC}_{\mathrm{REC}}$	Hyperparams
VICReg + corruption	0.819 ± .004	3 loss weights + EMA
NaiveJEPA	**0.779**	**1 (EMA only)**

Encoder Ranking — Simple Conv Wins

Encoder	Type	Params	$BACC_{REC}$
Conv1D	Strided conv	0.4M	**0.819**
BIOT-style	FFT patches	1.2M	0.790
LaBraM-style	Raw patches	1.2M	$\sim$0.775[†]
EEGPT	Ch-pool Tr	0.8M	collapsed

[†] accuracy, not BACC (early eval)

Why Conv wins:

The 0.4M conv has no architectural mismatch with corruption augmentation, trains stably, and plateaus at 0.82 regardless of depth, width, or data volume.

EEGPT failure: architectural conflict

- Cross-attention pools 19 channels → 1 token per time step
- Channel-drop creates views with *different channel sets*
- The two views are structurally incompatible before the encoder sees them
- Fix: remove channel-drop → TUEV 0.182 → 0.358

Lesson: always ablate augmentations *per* encoder family.

Honest Position vs Literature (Per-Window)

The evaluation trap: per-window vs per-recording (≈5 pp gap)
Literature (BIOT, LaBraM, FEMBA) reports **per-window** BACC. Our evaluation mean-pools 16 windows/patient → **per-recording**, which is 5 pp higher for the same encoder. Comparing them manufactures a false win.

Method	BACC_{WIN}	AUROC
ContraWR (SSL)	0.775	0.846
BIOT pretrained	0.802	0.874
LaBraM-Base	0.814	0.902
EB-JEPA SIGReg + corrupt	**0.775**	**0.856**
EB-JEPA VICReg + corrupt	0.770	0.848
EEGNet (supervised)	0.796	0.882

Honest reading:

- ✓ Matches ContraWR (1.6M) with 0.4M parameters
- × 3.9 pp below LaBraM-Base (5.8M, 200 ep)
- Competitive with EEGNet (supervised) as a *frozen* representation
- Per-recording our best is 0.825 / fine-tuned 0.812 (ties supervised EEGNet)

Cross-Task Transfer to TUEV (6-class Event Detection)

Protocol: freeze TUAB-pretrained encoder; train linear probe on TUEV (no TUEV labels during SSL).

Method	TUEV BACC	Trained on TUEV?
ST-Transformer	0.3385	yes, full
CNN-Transformer	0.4100	yes, full
VICReg (TUAB SSL)	**0.382**	probe only
NaiveJEPA (TUAB SSL)	**0.389**	probe only
FFCL	0.4551	yes, full
ContraWR	0.4664	yes, full
BIOT pretrained	0.5009	yes, full

Key point:

Our encoder **never saw a single TUEV label**. It was pretrained on TUAB binary pathology labels.

A linear probe on top beats **ST-Transformer and CNN-Transformer** trained end-to-end on TUEV.

Transfer from a binary task to a 6-class task — with only a logistic regression — is the strongest evidence that the SSL representation is genuinely general.

Summary

What works

- **Corruption augmentation**: $+2.3$ pp BACC (3-seed, robust)
- $\lambda_{\text{inv}}=1$: beats paper-correct $\lambda=25$ by 3 pp/6σ — EEG-specific finding
- **SIGReg** $\geq$ **VICReg**: consistent $+0.5$–1 pp AUROC
- **NaiveJEPA**: 0.779 BACC with 1 hyperparameter (no anti-collapse term)
- **TUEV transfer**: beats two supervised baselines without TUEV labels

What doesn't

- Scaling the encoder (capacity is not the limit)
- $4\times$ pretraining data (conv saturates)
- EEGPT (channel-drop $\times$ channel-pool conflict)
- Spectral auxiliary loss (neutral, not beneficial)
- GRU predictive JEPA (same-window prediction degenerates)

Open next step

λ_{inv} sweep $\{1, 5, 10, 25\}$ running now. If monotone ordering confirms, EEG SSL prefers weaker invariance — first systematic evidence in the literature.

Backup — Per-Recording Table (Clinical Metric)

Model	BACC_{REC}	AUROC	Eval
Conv + VICReg + corruption (3 seeds)	**0.819 ± .004**	**0.900 ± .006**	Frozen
Conv + SIGReg + corruption	0.825	0.913	Frozen
Conv + NaiveJEPA	0.779	0.863	Frozen
Conv + VICReg + corruption (FT, 3 seeds)	0.812 ± .004	0.908 ± .006	Fine-tuned
EEGNet (supervised)	0.812	0.882	—
LaBraM-Base (per-win only)	0.814_{WIN}	0.902_{WIN}	FT

Per-recording = mean-pool 16 windows/patient. Not directly comparable to per-window literature. Best per-window: 0.775 (SIGReg + corrupt). The 0.836 result (spectral-0.1, single-seed, eval-selected coefficient) is confounded and not reported here.

Backup — VICReg λ_{inv} Ablation (3 seeds each)

λ_{inv}	BACC_{REC}	AUROC	Note
1 (default)	$\mathbf{0.819 \pm .004}$	$\mathbf{0.900 \pm .006}$	anti-collapse dominates
5	running	—	capDinv5
10	running	—	capDinv10
25 (paper)	$0.789 \pm .005$	$0.882 \pm .004$	invariance dominates

Gap $\lambda = 1$ vs $\lambda = 25$: 3 pp, $\approx 6\sigma$ — not noise.